

# Hossein ZareMehrjerdi

PhD Candidate in Computer Engineering | AI/ML, LLMs, Computer Vision | Agentic AI

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## PROFESSIONAL SUMMARY

PhD Candidate in Computer Engineering specializing in Machine Learning, Large Language Models (LLMs), Computer Vision, and AI Systems Engineering. Experienced in designing and deploying production-grade AI systems, including Text-to-SQL platforms and multi-agent architectures. Strong background in Python, PyTorch, SQL, cloud platforms (AWS, Azure, GCP), and MLOps. Proven ability to deliver scalable AI solutions that improve workflow efficiency and enable non-technical users to interact with complex data systems.

## WORK EXPERIENCE

### Data Science & Visualization Intern

May 2025 – Aug 2025

Bayer, Chesterfield, MO

- Built AI-driven image analysis pipelines for Cell Painting to accelerate pesticide discovery and compound screening.
- Workflow Democratization:** Reduced workflow time from **1 week to 30 minutes** by designing cloud infrastructure (AWS, Docker) with automated provisioning, democratizing access across non-technical scientists.
- Foundation Models:** Deployed foundation-model-based pipelines (DINO, OpenFold) to extract molecular-level features, accelerating hypothesis generation for downstream biological experiments.

### Data Scientist Intern

May 2024 – Aug 2024

Corteva Agriscience, Des Moines, IA

- Launch LLM (Text-to-SQL):** Engineered a production Text-to-SQL system on Azure, translating natural language to SQL queries with advanced prompt engineering and RAG; achieved **~70% query accuracy** on internal benchmarks, enabling self-serve data access for non-technical stakeholders.
- Built and maintained secure data pipelines, ensuring controlled access to enterprise data without direct database exposure.
- Corteva Chat:** Deployed an internal LLM-based enterprise chat solution on Azure to **1,000+ users** via CI/CD pipelines, ensuring scalable and secure rollout across business units.

### Graduate Research Assistant

Jan 2023 – Present

Iowa State University, AI Institute for Resilient Agriculture (AIIRA), Ames, IA

- Ag-LLM:** Engineered a RAG pipeline over large-scale domain corpora powering a live AI pest identification system ([pest-id.las.iastate.edu](http://pest-id.las.iastate.edu)), answering farmer queries across hundreds of pest species.
- Remote Sensing & Phenotyping:** Built computer vision pipelines on UAV-based RGB imagery and canopy height data to automate trait extraction at field scale, significantly reducing expert annotation effort.
- Research & Publications:** Co-authored 6+ publications in AI/ML, collaborating with plant scientists, biologists, and domain experts across multiple institutions.

## SELECTED PROJECTS

### Bio-Shield — Agentic AI Biosecurity System

2025 – Present

- Designing a multi-agent system for real-time identification and risk assessment of invasive insects and pests deemed dangerous to U.S. crops.
- Agentic Architecture:** Orchestrating agentic workflows using LangGraph and LangChain, connecting specialized sub-agents via tool-calling and MCP servers.

### AgReason — LLM Reasoning Benchmark & Fine-Tuning for Agriculture

2025

- Co-developed the **first benchmark** for evaluating reasoning models on agriculture-specific tasks.
- Dataset Curation:** Built a custom omics-grounded reasoning dataset, integrating structured biological data with scientific literature to support multi-step inference.
- Model Fine-Tuning:** Fine-tuned reasoning models using **PEFT** (parameter-efficient fine-tuning) to adapt them to domain-specific tasks at low cost ([project page](#)).

## City of Des Moines (Capital of Iowa) — Automated Property Assessment with VLMs Sep 2025 – Present

- Designed an automated property assessment pipeline using Vision-Language Models for large-scale municipal infrastructure condition evaluation across the city.
- **Multi-Model Consensus:** Achieved **93% accuracy** by aggregating predictions across multiple VLMs; reduced turnaround from months to days and cut evaluation cost to **one-tenth** of the manual process.
- **Executive Presentation:** Presented findings directly to the City Manager of Des Moines, translating complex AI methodology into actionable insights for municipal leadership.

## MaizeEar-SAM — Computer Vision · Zero-Shot Phenotyping

2023 – 2025

- Developed a zero-shot maize ear phenotyping pipeline using Segment Anything Model (SAM), eliminating costly manual annotation at scale.
- Implemented graph-based kernel counting; code at [github.com/mhzarem/MaizeEar-SAM](https://github.com/mhzarem/MaizeEar-SAM).

## Large-Scale Multi-Modal Molecular Representation Learning — Research Project

2024 – 2025

- **HPC Pipeline:** Built an HPC pipeline generating 3D molecular geometries for **2.8 billion molecules**, enabling large-scale pre-training of molecular foundation models.
- **Multi-Modal Foundation Model:** Developed a multi-modal embedding model fusing 1D SMILES, 2D graphs, and 3D conformers, improving downstream molecular property prediction.

## EDUCATION

Iowa State University, Ames, IA

Aug 2023 – Dec 2026 (expected)

PhD Candidate, Computer Engineering | GPA: 4.0/4.0 | Focus: AI/ML, Computer Vision, LLMs

Iowa State University, Ames, IA

2023 – 2024

M.S., Computer Engineering | GPA: 4.0/4.0

Amirkabir University of Technology, Tehran, Iran

2015 – 2020

B.S., Computer Engineering | GPA: 3.8/4.0 | Thesis: Implementation of CNN with CUDA

## TECHNICAL SKILLS

**Core AI Domains:** Computer Vision, Natural Language Processing (NLP), Large Language Models (LLMs), Vision-Language Models (VLMs), Deep Learning, Multi-Modal Learning

**Agentic AI & Orchestration:** LangChain, LangGraph, MCP Servers, Multi-Agent Systems, RAG, RAGAs, LangSmith

**AI/ML Frameworks:** PyTorch, Hugging Face Transformers, OpenCV, Scikit-learn

**Foundation & Frontier Models:** Llama, Mistral, SAM, DINO, OpenFold, PEFT/LoRA fine-tuning

**Cloud & MLOps:** AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Git

**Programming Languages:** Python, C++, C, Java, JavaScript, R, PHP, SQL, MATLAB, Bash

**Databases & Vector Stores:** PostgreSQL, MongoDB, ChromaDB, MySQL

**Web & APIs:** FastAPI, React, Django, Flask, REST APIs

## PUBLICATIONS

- **ZareMehrjerdi, H.**, Ganguly, S., Rairdin, A., et al. “Towards Large Reasoning Models for Agriculture.” arXiv:2505.19259, 2025.
- **ZareMehrjerdi, H.**, Coffey, L., Jubery, T., et al. “MaizeEar-SAM: Zero-Shot Maize Ear Phenotyping.” arXiv:2502.13399, 2025.
- Chiranjeevi, S., **ZareMehrjerdi, H.**, et al. “TerraIncognita: A Dynamic Benchmark for Species Discovery Using Frontier Models.” arXiv:2506.03182, 2025.
- Singh, A. K., Jones, S. E., **ZareMehrjerdi, H.**, et al. “Use of Artificial Intelligence in Soybean Breeding and Production.” Advances in Agronomy, 2025.
- Hamidi Rad, R., **ZareMehrjerdi, H.**, et al. “Recommendation System over Multi-layer Complex Networks.” SAI Intelligent Systems Conference, Springer, 2023.